

Multiple Choice

1. **Answer:** B

Options: A) Liver glycogen; B) Muscle glycogen; C) Intramuscular lipid; D) Adipose tissue lipid

Note: Muscle glycogen is the primary source of energy during moderate to high-intensity exercise because it is readily available for rapid conversion into glucose, which fuels muscle contractions.

2. **Answer:** A

Options: A) Cows' milk; B) Natural unsweetened orange juice; C) Honey; D) Agave nectar

Note: Cows' milk does not contain free sugars because it primarily consists of lactose, which is a naturally occurring sugar found in dairy products and is not classified as a free sugar.

3. **Answer:** B

Options: A) Niacin; B) Riboflavin; C) Thiamin; D) Vitamin C

Note: Riboflavin, also known as vitamin B₂, is essential for the production of flavin adenine dinucleotide (FAD), which is a crucial cofactor in the beta-oxidation pathway that breaks down fatty acids for energy.

4. **Answer:** C

Options: A) Glucagon increases the rate of glycolysis; B) Glycolysis requires NADP⁺; C) In glycolysis, glucose is cleaved into two three-carbon products; D) The end-product of glycolysis in red blood cells is pyruvate

Note: The correct answer is accurate because glycolysis is the metabolic pathway that breaks down one molecule of glucose, a six-carbon sugar, into two molecules of pyruvate, which are three-carbon compounds.

5. **Answer:** A

Options: A) Synthesis of TMP (thymidine monophosphate); B) Decarboxylation of amino acids to form amine neurotransmitters; C) Synthesis of methionine from homocysteine; D) Carboxylation of pyruvate to oxaloacetate

Note: Folate is essential for the synthesis of nucleotides, including thymidine monophosphate (TMP), which is crucial for DNA replication and repair.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The 'fear of becoming fat' in patients with anorexia nervosa is considered a

distorted perception of body image rather than a paranoid delusion, as it stems from a psychological disorder rather than a break from reality.

7. **Answer:** False

Options: A) True; B) False

Note: A baked apple has a higher glycaemic index than a raw potato because cooking increases the sugar availability in the apple, leading to a quicker rise in blood sugar levels compared to the more slowly digested carbohydrates in a raw potato.

8. **Answer:** False

Options: A) True; B) False

Note: Older adults typically experience a decrease in muscle mass due to age-related factors such as hormonal changes, decreased physical activity, and loss of muscle fibers.

9. **Answer:** False

Options: A) True; B) False

Note: Muscle contains more water than fat, which actually facilitates the travel of electrical currents rather than slows it down.

10. **Answer:** False

Options: A) True; B) False

Note: High sodium intake is associated with increased blood pressure and can exacerbate hypertension rather than improve it.

Long Form Response

11. **Answer:** The relationship between dietary factors, particularly the consumption of salt and salt-preserved foods, and the increased risk of stomach cancer is significant. High salt intake can damage the stomach lining and promote inflammation, which may contribute to cancer development. Additionally, salt-preserved foods, such as pickles and cured meats, often contain harmful substances that can increase cancer risk. Studies have shown that populations with high salt consumption tend to have higher rates of stomach cancer. Therefore, reducing salt intake and limiting salt-preserved foods can be important steps in lowering the risk of this disease.
Note: correct because it identifies the damaging effects of high salt intake on the stomach lining, the inflammatory response it triggers, and the association of salt-preserved foods with harmful substances that collectively contribute to an increased risk of stomach cancer.

12. **Answer:** The Dutch Hunger Winter of 1944-1945 had a significant impact on the DNA methylation of individuals who were exposed to starvation while in utero, especially during the first trimester of pregnancy. During this critical period, the developing fetus is highly sensitive to environmental factors, including nutritional deficiencies. Research shows that those exposed to malnutrition early in pregnancy experienced changes in DNA methylation patterns, which can affect gene expression and influence health in adulthood. This altered methylation can lead to increased risks of various conditions,

such as cardiovascular diseases and metabolic disorders. Thus, the first trimester is particularly significant as it sets the foundation for how genes are regulated later in life.

Note: correct because it highlights the vulnerability of the developing fetus to environmental factors like malnutrition during the first trimester, which has been shown to lead to lasting changes in DNA methylation and gene expression in adult offspring.

End of Answer Key